# Enclosure & Home-Run Install Packet

Companion to the bench build. Covers the parts, cabling, terminations, and field practices to move the monitor from breadboard into the trailer with remote (long home-run) door and leak sensors on a dedicated 5V USB-C feed.

## Scope & assumptions

| Parameter | Value / result |
|---|---|
| Door contact home run | 25 ft |
| Leak sensor home run | 15 ft |
| USB-C power run | 6 ft |
| Pull slack + service loops | +14% and 2 ft/end |
| Signal cable needed | ~54 ft -> buy 100 ft spool 22/4 shielded |

Adjust the run lengths at the top of the generator and all footage recalculates.

## 1. Install BOM addendum

Adds to the base bill of materials; does not replace it. [FUTURE] items are optional.

| Qty | Item | Spec / rating | Purpose | Notes |
|---|---|---|---|---|
| 1 | Enclosure, polycarbonate NEMA 4X (non-metallic) | ~8x6x4 in, IP66/4X, clear or opaque lid | House board + DIN terminals; RF-transparent for WiFi | MUST be non-metallic - a metal box blocks the on-board WiFi antenna. |
| 1 | DIN rail, 35 mm | steel/aluminum, cut to ~6 in | Mount terminal blocks + board carrier | Trim to enclosure interior width. |
| 10 | Feed-through terminal blocks, 2.5 mm^2 | screw or spring, DIN-mount | Land all field conductors (serviceable) | 7 used + spares. Do NOT solder field wires to the board. |
| 1 | Terminal end bracket + end plate set | 35 mm DIN | Retain the terminal stack | 2 end brackets + 1 end plate typical. |
| 2 | Ground/shield terminal block | grounding type, DIN-mount | Single-point shield/GND landing bar | Shield lands here at ENCLOSURE end only. |
| 3 | Cable gland, IP68 | PG7 or M12, 3.0-6.5 mm cable OD | Seal + strain-relief the two signal home runs | 2 used (door, leak) + 1 spare. Bottom entry. |
| 1 | USB-C panel-mount bulkhead pass-through | IP67, C-female to C-female | Power entry for the dedicated 5V supply | Cleaner + sealed vs. trying to gland a molded USB plug. |
| 1 | Shielded cable, 22 AWG / 4-cond, stranded (100 ft spool) | overall foil shield + drain, ~5-6 mm OD | Both home runs (standardized on one SKU) | ~54 ft needed (door 25 + leak 15 + slack/loops). 22/4 gives a spare conductor for future use. |
| 1 | Wire ferrules, 22 AWG (0.34 mm^2) + crimp tool | insulated ferrules assortment | Terminate stranded conductors into terminals | Matches Celestial ferrule discipline; prevents strand splay. |
| 2 | Resistor, 4.7k ohm 1/4 W | 5% carbon film | External pull-up: door line D2 -> 5V | 1 used + spare. Stiffens the long door line vs. noise. |
| 2 | Panel-mount LED holder/bezel, 5 mm | chrome or black bezel | Face-mount the green/red status LEDs | Uses the 5mm LEDs + 220 ohm from the base BOM. |
| 4 | Anti-vibration mount / rubber isolator | well-nut or grommet mount, M4/M5 | Isolate enclosure from trailer vibration | Prevents fatigue on solder joints + terminals. |
| 1 | Self-fusing silicone tape + heat-shrink assortment | - | Shield termination + far-end drain dressing | Cut + tape shield/drain at each sensor end. |
| 1 | Wire label set (heat-shrink or self-lam wrap) | printable | Both-end labels per pull schedule | Label DR-01, WL-01, PWR-01, ENC-01. |
| 1 | Cable ties + adhesive mounts | UV-rated (trailer) | Dress conductors + form drip loops | Use UV-rated for a mobile/outdoor environment. |

| Qty | Item | Spec / rating | Purpose | Notes |
|---|---|---|---|---|
| 1 | [FUTURE] Waterproof DS18B20 1-Wire temp probe + 4.7k | 1-Wire, waterproof lead | Remote temperature WITHOUT extending I2C | Optional. I2C/BME280 must stay local; 1-Wire runs long. |
| 2 | [FUTURE] Clamp-on ferrite core | for 5-6 mm cable | Extra EMI suppression on home runs if needed | Optional. Add only if noise persists after shielding. |

## 2. Pull / label schedule - field home runs

| Tag | From | To | Cable | Cond. used | Len | Notes |
|---|---|---|---|---|---|---|
| DR-01 | ENC-01 TB (D2 sig + GND) | Door contact (frame) | 22/4 shielded | white=sig, black=return (2 of 4) | 25 | Reed on frame, magnet on leaf. Shield to GND bar at ENC end only; cut+tape at device. Drip loop at gland. |
| WL-01 | ENC-01 TB (D3 DO, 5V, GND) | Leak sensor (floor/drip pan) | 22/4 shielded | red=V+, black=GND, white=DO (3 of 4) | 15 | Mount at lowest point. Green = spare. Shield at ENC end only; cut+tape at device. Drip loop at gland. |
| PWR-01 | 5V USB-C supply (dedicated) | ENC-01 USB-C bulkhead | USB-C (supply lead) | USB-C | 6 | >= 2 A supply. Enter via IP67 bulkhead, NOT a gland. Short/quality cable; drip loop. |

## 3. Internal termination map (enclosure)

| Board pin | Terminal | Field conductor | Notes |
|---|---|---|---|
| D2 | TB-1 (DR sig) | DR-01 white | Door signal; 4.7k pull-up TB-1 -> 5V |
| GND | TB-2 (DR ret) | DR-01 black | Door return |
| D3 | TB-3 (WL DO) | WL-01 white | Leak digital out |
| 5V | TB-4 (WL V+) | WL-01 red | Leak sensor power |
| GND | TB-5 (WL GND) | WL-01 black | Leak return |
| - | GND/shield bar | both drains | Shields land here, ENC end ONLY |
| D5 | via 220 ohm | green LED | Face LED - NORMAL (internal) |
| D6 | via 220 ohm | red LED | Face LED - ALERT (internal) |
| USB-C | board USB-C port | bulkhead | Power from PWR-01 bulkhead |

## 4. Field practices (why)

- **Non-metallic enclosure.** The UNO R4 WiFi uses a PCB trace antenna; a metal box is a Faraday cage and will silently kill the Cloud link. Polycarbonate NEMA 4X keeps RF and the moisture/corrosion rating.
- **Shield single-point ground.** Land the drain at the enclosure GND bar only; cut and tape it at each sensor. Grounding both ends creates a loop that injects noise instead of blocking it.
- **Door pull-up.** The internal INPUT_PULLUP (~20-50k) is too weak for a long line. A 4.7k to 5V stiffens it against EMI false-trips. The leak sensor output is actively driven, so it needs only the shield.
- **Keep BME280 local.** I2C is not a long-run bus. For remote temperature use a DS18B20 (1-Wire), which is built for distance.
- **Power margin + entry.** Feed the board USB-C from a dedicated >= 2 A supply through an IP67 bulkhead (not a gland). Undersized supplies brown out on WiFi bursts.
- **Mechanical.** Bottom-entry glands with drip loops; land field wires on DIN terminals (never soldered to the board); mount the enclosure on vibration isolators.
- **Firmware backstop (REV 2.0).** Door/leak inputs require consecutive agreeing samples before a state change, and the BME280 self-recovers from a dropped connection - both matter on a vibrating, electrically noisy trailer.